

Fig. 9: E-tongue software screen



Fig. 10: E-tongue sensors

conversion of the membrane response into signals that will be analysed

Circuit level. At the circuit level, the sample is quantified and digitised, and results recorded. Electro-analytical methods can be used for measuring the result. These include both active and passive to study an analyte through the observation of potential and/or current.

Potentiometry measures the potential of a solution between two electrodes simply, without affecting the solution much in the process. Voltammetry applies a constant and/or varying potential at an electrode surface and measures the resulting current with a three-electrode system. Coulometry uses applied current or potential to completely convert an analyte from one oxidation state into another. In this experiment, the total current passed is measured directly or indirectly to determine the number of electrons passed

Perceptual level. Perception is done in the computer using the e-tongue system. Depending upon the application for which it

is applied, the data analysis can produce a variety of information. Exploratory data analysis can be performed using different statistical methods such as principal component analysis and linear discriminant analysis on the data set to correlate the collected data.

Tea tasters usually evaluate tea quality and conventionally assign scores on a scale of 1 to 10 depending on the major quality attributes of the tea sample. The major quality attributes of black tea are briskness/strength, flavour, aroma and colour.

At C-DAC Kolkata, e-tongue for determining the briskness/strength of the black tea liquor has been developed. The virtual instrumentation-based e-tongue uses voltammetry technique for correlating the briskness/strength of tea liquor with the 'tea taster' mark. In this work, an array of selected electrodes was immersed into black tea liquor-samples for analysing the tea samples. A computational model was developed to correlate the measurements with the tea taster's briskness/strength scores. The developed e-tongue structure is shown in Fig. 8.

4 GSM-based remote operation of agricultural equipment

Bengaluru based Rural Bridges' Kisan Shakti is an easy-to-use and economical communication device which enables a farmer to operate his motor pump set from anywhere through his mobile or landline. The GSM-based controller can be fitted to all the existing motor starters and used with mobiles or landlines of any operator.

How it works? A farmer simply calls Kisan Shakti and selects the options provided by the interactive voice response system (IVRS) menu in the local language to switch on/off of the motor. He gets the confirmation of the operation performed along with running status of the motor and also if water is being pumped or not immediately from the IVRS menu. Advanced technological features of the device ensure instant communication to the farmers whenever power is available, and also in cases of faulty power supply, lack of water in the well/bore and attempt of

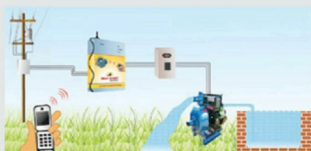


Fig. 11: Working of Kisan Shakti

device theft. The device raises an alarm and makes a phone call to the farmer if anyone tries to steal it or any device connected to it. There are different modes of operation to suit the needs of the farmers—auto, manual and timer—indicated by three LED indicators. The device is adaptive to any motor configuration up to 20 HP.